EXPERIMENT-1

Title

Design an ER Diagram for an Indian E-Commerce Platform

Aim

To design an Entity Relationship (ER) Diagram for an Indian E-Commerce Platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, and Address, including primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.

Theory

An Entity Relationship (ER) Model is a conceptual representation of data used in database design. It describes entities, their attributes, and relationships among them.

Key Components

1. **Entity** – Real-world object represented by a rectangle.
2. **Attribute** – Property of an entity represented by an oval.
3. **Primary Key** – Uniquely identifies an entity.
4. **Composite Attribute** – Attribute consisting of multiple sub-attributes.
5. **Multi-valued Attribute** – Attribute having multiple values.
6. **Weak Entity** – Entity that depends on another entity for identification.
7. **Relationship** – Association between entities.
8. **Specialization** – Dividing an entity into subclasses.
9. **Participation Constraint** – Specifies mandatory or optional participation.

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ER DIAGRAM:

